Lujo Hemorrhagic Fever: Comprehensive Disease Characterization

Disease: Lujo Hemorrhagic Fever (LHF) · **MONDO:** MONDO:0017872 · **Synonym:** Zambian hemorrhagic fever · **Category:** Infectious (viral hemorrhagic fever)

Summary

Lujo hemorrhagic fever is an infectious disease, not a Mendelian genetic disorder — it has no human causal gene. Its causal agent is **Lujo virus (LUJV)**, a genetically divergent Old World mammarenavirus (family *Arenaviridae*, genus *Mammarenavirus*) first identified during a 2008 nosocomial outbreak in Southern Africa. The template field "Causal gene(s): not recorded" is therefore correct and expected: the relevant "gene" is viral, not host. LUJV is an enveloped, bisegmented, negative-sense/ambisense RNA virus whose full-genome phylogeny places it at the ancestral node of the Old World arenaviruses, almost equidistant from both the Old World (Lassa) and New World serogroups (PMID: 19478873).

The entire recognized clinical experience with LHF derives from **a single 2008 outbreak of five cases** (an index patient air-evacuated from Lusaka, Zambia, to Johannesburg, South Africa, plus four healthcare workers infected through nosocomial transmission), with a **case fatality rate of 80% (4 of 5)**. The disease begins as a nonspecific febrile prodrome and progresses over roughly two weeks to a capillary-leak syndrome, disseminated intravascular coagulation (DIC), shock, and multi-organ failure — a course clinically indistinguishable from severe Lassa fever. Notably, overt major hemorrhage is uncommon despite the "hemorrhagic fever" label.

Mechanistically, LUJV is distinguished from all other arenaviruses by its **unique cell-entry pathway**: it uses **neuropilin-2 (NRP2)** as its primary cell-surface receptor and the tetraspanin **CD63** (critically, residue phenylalanine-143 in humans) as an intracellular fusion cofactor in the late endosome — a two-receptor "switch" paradigm analogous to Lassa virus's α-dystroglycan→LAMP1 switch. Its glycoprotein precursor (GPC) is matured by host **site-1 protease (S1P/SKI-1/MBTPS1)** at an RKLM motif, and the virus buds via Z-protein late domains independently of the canonical ESCRT machinery. Disease severity is driven primarily by **host**

immune dysregulation (cytokine storm) rather than viral burden. There is **no approved vaccine or antiviral**; management rests on early ribavirin plus intensive supportive care and strict infection control, with numerous candidate countermeasures (CB1-antagonist entry inhibitors, the S1P inhibitor PF-429242, griffithsin, and the MOPEVAC vaccine platform) remaining preclinical.

Key Findings

Finding 1 — LHF is caused by Lujo virus, a divergent Old World arenavirus; there is no human causal gene

Lujo virus is the etiologic agent of LHF. It is an enveloped, bisegmented negative-sense/ambisense RNA virus of the family *Arenaviridae*, genus *Mammarenavirus*. Full-genome analysis during its discovery showed LUJV to be "unique and branching off the ancestral node of the Old World arenaviruses," with a G1 glycoprotein sequence "highly diverse and almost equidistant from that of other Old World and New World arenaviruses" (PMID: 19478873). This places LUJV outside both classical arenavirus serogroups. The name "Lujo" is a portmanteau of **Lu**saka and **Jo**hannesburg, the two cities linked by the 2008 outbreak. Because the disease is infectious, the "causal gene" fields (OMIM, HGNC, causal gene) are not applicable; the disease-level identifier is **MONDO:0017872**.

Evidence quote: *"Full genome analyses of LUJV showed it to be unique and branching off the ancestral node of the Old World arenaviruses. The virus G1 glycoprotein sequence was highly diverse and almost equidistant from that of other Old World and New World arenaviruses"* (PMID: 19478873).

Finding 2 — LUJV uses NRP2 as entry receptor and CD63 as intracellular fusion cofactor, distinct from all other arenaviruses

A genome-wide haploid genetic screen identified the transmembrane protein **neuropilin-2 (NRP2)** and the tetraspanin **CD63** as the host factors required for LUJV glycoprotein (GP)-mediated infection. LUJV GP1 binds the **N-terminal (a1/a2) domain of NRP2**, while CD63 stimulates pH-activated LUJV GP-mediated membrane fusion in the late endosome (PMID: 29120745). This is mechanistically distinct from Old World arenaviruses (which use α-dystroglycan) and New World arenaviruses (which use transferrin receptor 1); pseudotype

studies confirmed LUJV entry is independent of both α-dystroglycan and hTfR1, and depends instead on an unidentified receptor later shown to be NRP2, with additional dependence on Niemann-Pick C1 (NPC1) and cholesterol transport (PMID: 24741091). Structural work defined the receptor-binding mode by X-ray crystallography of the GP1–NRP2 complex (PMID: 30150732), and cryo-EM of the complete trimeric spike revealed a single NRP2 molecule bound at the apex via an unusual arginine-methionine interaction (PMID: 39169025). The host-range determinant is a **phenylalanine at position 143 in the human CD63 large extracellular loop**, which is required for efficient membrane fusion and infection (PMID: 35164564).

Evidence quotes: *"A genome-wide haploid genetic screen identified the transmembrane protein neuropilin 2 (NRP2) and tetraspanin CD63 as factors for LUJV GP-mediated infection. LUJV GP binds the N-terminal domain of NRP2, while CD63 stimulates pH-activated LUJV GP-mediated membrane fusion"* (PMID: 29120745); *"a phenylalanine at position 143 in human CD63 was the key residue for efficient membrane fusion and VSVΔG*-LUJV/GP infection"* (PMID: 35164564).

Finding 3 — Clinical course: febrile prodrome progressing to shock, DIC, and multi-organ failure

In the 2008 outbreak, illness "typically began with the abrupt onset of fever, malaise, headache, and myalgias followed successively by sore throat, chest pain, gastrointestinal symptoms, rash, minor hemorrhage, subconjunctival injection, and neck and facial swelling over the first week of illness. No major hemorrhage was noted." Neurological signs appeared late. "Shock and multi-organ system failure, often with evidence of disseminated intravascular coagulopathy, ensued in the second week, with death in four of the five cases." The single survivor received early ribavirin plus aggressive supportive care (PMID: 25393244). Laboratory features closely mirror Lassa fever — fever, thrombocytopenia, and transaminitis — and the clinical syndrome is "remarkably similar to Lassa hemorrhagic fever," with fever, headache, vomiting, diarrhea, arthralgia, myalgia, and signs of hemorrhagic syndrome (PMID: 29733163).

Finding 4 — Host immune response, not viral burden, drives LHF severity

Two independent animal models converge on the conclusion that **host immune dysregulation determines outcome**. In cynomolgus macaques, LUJV caused mild non-lethal illness, and transcriptomics of PBMCs revealed a **72-hour delay** in the induction of inflammatory, antiviral, and cell-death gene programs compared with lethal Lassa infection (PMID: 25520505). In strain 13/N guinea pigs, survival correlated with the immune response rather than viral titer: despite similarly high LUJV titers across lungs, liver, spleen, kidneys, and serum, young females survived with low inflammatory cytokines, whereas adults and

young males died with cytokine storms featuring elevated IL-1β, IL-2, IL-6, and CXCL10 (PMID: 41118368). This reveals both a central pathogenic mechanism (immunopathology) and age/sex disease modifiers.

Evidence quotes: *"survival outcomes strongly correlated with immune responses rather than viral burden. Adult animals and young males developed more severe clinical signs, heightened hematological and biochemical abnormalities, and pronounced cytokine storms, particularly elevated IL-1β, IL-2, IL-6, and CXCL10 levels"* (PMID: 41118368); *"We detected a 72-h delay in the induction of host responses to infection during LUJV infection compared to that of the animals infected with LASV... particularly apparent among groups of genes promoting cell death"* (PMID: 25520505).

Finding 5 — Animal models: macaque (mild), guinea pig and STAT-1 KO mouse (partial lethal disease)

No single model fully reproduces fatal human LHF, but complementary systems exist. Cynomolgus macaques develop mild, non-lethal illness (PMID: 25520505). Strain 13/N guinea pigs are "a promising model for LUJV pathogenesis," recapitulating hematologic/biochemical abnormalities and age/sex-dependent lethality (PMID: 41118368). STAT-1-deficient mice develop a partially lethal disease in which "several key hematological, clinical chemistry, and histologic findings common to LHF disease are recapitulated" (PMID: 42075721). Standard inbred mice and hamsters are relatively resistant because their CD63 lacks the human F143 residue required for efficient entry (PMID: 35164564). Pseudotyped VSV bearing LUJV GP enables BSL-2 study of entry (PMID: 24741091; PMID: 22396728).

Finding 6 — Epidemiology: a single 2008 outbreak, 80% CFR, reservoir unknown, no approved countermeasures

"Lujo virus is a novel Old World arenavirus identified in Southern Africa in 2008 as the cause of a viral hemorrhagic fever (VHF) characterized by nosocomial transmission with a high case fatality rate of 80% (4/5 cases)" (PMID: 27593704). The index patient from Lusaka, Zambia, was air-evacuated to Johannesburg, with secondary transmission to four healthcare workers. LUJV remains the only strain detected to date (PMID: 24096270), and "LUJV remains poorly understood, with no approved treatments, vaccines, or known reservoir" (PMID: 41118368). An arenaviral rodent reservoir is presumed but unidentified. Rapid RT-PCR assays were developed for molecular diagnosis (PMID: 24096270). LUJV is a BSL-4 / WHO risk-group-4 pathogen.

Finding 7 — GPC matured by site-1 protease at an RKLM motif; Z-protein drives ESCRT-independent budding

LUJV's glycoprotein precursor GPC is proteolytically cleaved by cellular **site-1 protease (S1P/SKI-1, gene MBTPS1)** at an **RKLM motif**, and the S1P inhibitor PF-429242 reduced LUJV production. The matrix protein Z **late (L) domains are required for efficient virus-like-particle release, but the virus buds independently of the canonical ESCRT factors Tsg101, ALIX/AIP1, and Vps4A/B** (PMID: 26719243). Both the S1P maturation step and the Z-driven budding step represent druggable nodes.

Evidence quotes: *"LUJV GPC is cleaved by site 1 protease (S1P) at the RKLM motif, and treatment with the S1P inhibitor PF-429242 reduced LUJV production"* and *"The L domains in Z are required for efficient virus-like particle release, but Tsg101, ALIX/AIP1, and Vps4A/B are unnecessary for budding"* (PMID: 26719243).

Finding 8 — Multiple experimental anti-LUJV strategies; only ribavirin used clinically

Using VSV-LUJV/GP pseudotype entry screens, **cannabinoid receptor 1 (CB1) antagonists** rimonabant, AM251, and AM281 were identified as robust inhibitors of LUJV entry (PMID: 37633192). FDA-approved drug-library screens identified additional entry inhibitors (PMID: 34925303). A recombinant reporter-LUJV high-throughput platform identified six compounds with robust anti-LUJV activity, including agents with prior pan-arenavirus activity (PMID: 34203149). The broad-spectrum lectin **griffithsin** blocks LUJV glycoprotein-mediated entry (PMID: 42037406). Clinically, only **ribavirin** (with supportive care) has been used, in the single survivor (PMID: 25393244). All targeted agents remain preclinical/investigational; none is approved for LHF.

Evidence quote: *"Cannabinoid receptor 1 (CB1) antagonists, such as rimonabant, AM251 and AM281, have been identified as robust inhibitors of LUJV entry"* (PMID: 37633192).

Finding 9 — Tissue injury follows shared VHF mechanisms; CD63 is LUJV's intracellular fusion cofactor analogous to LAMP1 for Lassa

Across arenaviral and filoviral hemorrhagic fevers, pathogenesis involves "deficiency of hepatic synthesis of coagulation factors owing to hepatocellular necrosis, cytokine storm, increased permeability by vascular endothelial growth factor, complement activation, and disseminated intravascular coagulation" (PMID: 23121052). In a primate Lassa model, shock arose from "biochemical dysfunctions of platelets and endothelial cells," with loss of

endothelial prostacyclin production and impaired platelet aggregation despite normal platelet counts (PMID: 3543155). LUJV parallels Lassa mechanistically: both are Old World arenaviruses whose GPC first engages a cell-surface receptor, then switches to an intracellular factor for low-pH fusion. Lassa switches from α-dystroglycan to LAMP1 (PMID: 29295909), whereas LUJV uses NRP2 then the tetraspanin CD63 (F143) (PMID: 29120745; PMID: 35164564).

Evidence quotes: *"The pathogenic mechanisms... include deficiency of hepatic synthesis of coagulation factors owing to hepatocellular necrosis, cytokine storm, increased permeability by vascular endothelial growth factor, complement activation, and disseminated intravascular coagulation"* (PMID: 23121052); *"A unique feature of LASV entry is a pH-dependent switch from a primary cell surface receptor (α-dystroglycan) to an endosomal receptor, lysosomal-associated membrane protein (Lamp1)"* (PMID: 29295909).

Finding 10 — NP exoribonuclease likely underlies delayed innate response; MOPEVAC is a candidate vaccine platform

Arenaviral nucleoproteins carry a conserved **DEDDH exoribonuclease (RNase) domain that suppresses type I interferon**; mutating the catalytic residues in Pichinde virus NP abolishes IFN suppression, causing high IFN induction and failure to replicate in IFN-competent cells and guinea pigs (PMID: 25878103). By homology, this conserved mechanism provides a molecular basis for the ~72-hour delay in innate/antiviral gene induction in LUJV-infected macaques (PMID: 25520505) and for the worsened disease in STAT-1-deficient mice (PMID: 42075721). For prevention, a recombinant hyperattenuated Mopeia virus platform (**MOPEVAC**) expressing heterologous arenaviral glycoproteins has been developed as a pan-arenavirus vaccine strategy (PMID: 29593043).

Evidence quote: *"Previous structural and functional analyses of arenavirus nucleoproteins (NPs) revealed a conserved DEDDH exoribonuclease (RNase) domain that is important for type I interferon (IFN) suppression"* (PMID: 25878103).

Report by Template Section

1. Disease Information

Lujo hemorrhagic fever is a severe, acute **viral hemorrhagic fever** caused by Lujo virus, an Old World mammarenavirus. The disease is defined at the disease level (not patient-EHR level); essentially all clinical knowledge derives from the aggregated five-case 2008 outbreak and subsequent experimental virology. - **Mondo:** MONDO:0017872 - **ICD-10:** A96.8 (other arenaviral hemorrhagic fevers); **ICD-11:** arenaviral hemorrhagic fever grouping (1D60.Y) - **OMIM:** Not applicable (infectious disease, no Mendelian locus) - **Orphanet:** Listed among rare arenaviral/viral hemorrhagic fevers - **MeSH:** "Hemorrhagic Fevers, Viral"; "Arenaviridae Infections" - **Synonyms:** Lujo viral hemorrhagic fever; Zambian hemorrhagic fever - **Pathogen taxonomy (NCBI Taxon):** Lujo mammarenavirus (taxid 649188); family Arenaviridae; genus Mammarenavirus

2. Etiology

Primary cause: infection with Lujo virus — an infectious, not genetic, etiology (PMID: 19478873). **Risk factors** are exposure-based: close/nosocomial contact with an infected patient or body fluids. Four of the five known cases were healthcare workers infected caring for the index patient (PMID: 27593704). No host genetic susceptibility or protective loci have been identified in humans. Experimental models reveal **age and sex as disease modifiers** (young females survived; adults and young males died in guinea pigs) (PMID: 41118368). Host **STAT1/type I IFN signaling is protective**, as STAT1-deficient mice develop lethal disease (PMID: 42075721). No gene–environment interaction data in humans exist (single small outbreak).

3. Phenotypes

Phenotype	Type	Suggested HPO	Frequency / notes
Fever	Symptom	HP:0001945	Universal, abrupt onset
Headache	Symptom	HP:0002315	Early prodrome
Myalgia	Symptom	HP:0003326	Early prodrome
Malaise/fatigue	Symptom	HP:0025406 / HP:0012378	Early
Sore throat (pharyngitis)	Sign	HP:0025439	Week 1
Chest pain	Symptom	HP:0100749	Week 1
Diarrhea / GI symptoms	Symptom	HP:0002014	Week 1
Skin rash	Sign	HP:0000988	Week 1
Subconjunctival hemorrhage/ injection	Sign	HP:0011885	Week 1
Facial/neck edema (swelling)	Sign	HP:0000316 (facial edema)	Distinctive late sign
Minor hemorrhage	Sign	HP:0001892	Present; major hemorrhage absent
Thrombocytopenia	Lab	HP:0001873	Lassa-like
Elevated transaminases	Lab	HP:0002910	Transaminitis
Shock	Sign	HP:0031273	Week 2, terminal
Disseminated intravascular coagulation	Lab/sign	HP:0005521	Week 2, terminal
Multi-organ failure	Sign	HP:0011024 + multiorgan	Week 2, terminal
Neurological signs	Sign	HP:0000707	Late

Onset is **adult** in the documented cases, **acute**, and **rapidly progressive** over ~2 weeks. Severity is severe/fatal in 4/5 cases. Quality-of-life measures are not applicable given the acute, high-fatality course. (Clinical features: PMID: 25393244; PMID: 29733163.)

4. Genetic/Molecular Information

Not applicable in the human-host genetic sense — **no causal human gene, OMIM entry, or pathogenic germline variant**. The relevant molecular biology is viral: the bisegmented genome (L and S segments) encodes the RdRp (L), matrix protein Z, nucleoprotein (NP), and glycoprotein precursor (GPC → GP1/GP2 + stable signal peptide). Host factors functionally required for the viral life cycle (effectively the human "molecular interactors") include **NRP2** (HGNC:8005), **CD63** (HGNC:1692; critical residue F143), **MBTPS1/S1P** (HGNC:15456), and **NPC1** (HGNC:7897) (PMID: 29120745; PMID: 35164564; PMID: 26719243; PMID: 24741091). No epigenetic or chromosomal abnormalities apply.

5. Environmental Information

The sole environmental/infectious factor is exposure to LUJV. The natural reservoir is **unknown** but presumed to be a rodent by analogy to other arenaviruses (PMID: 41118368). Documented human-to-human spread was **nosocomial**, via contact with blood/body fluids (PMID: 27593704). **Infectious agent:** Lujo mammarenavirus (NCBI taxid 649188).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating infection → clinical manifestation):

1. LUJV is introduced via mucosal/percutaneous exposure to infected body fluids, **leading to** infection of permissive cells (dendritic cells, macrophages, endothelium inferred by analogy to Lassa).
2. LUJV GP1 binds the N-terminal domain of cell-surface **NRP2**, **resulting in** receptor-mediated endocytosis (PMID: 29120745; PMID: 30150732).
3. In the acidifying late endosome, engagement of **CD63 (F143)** triggers pH-activated GP2-mediated membrane fusion (a receptor "switch" analogous to Lassa α-DG→LAMP1), **leading to** release of the ribonucleoprotein into the cytoplasm (PMID: 35164564; PMID: 29295909). *(GPC was first primed by host S1P cleavage at RKLM during biogenesis — PMID: 26719243.)*
4. Viral replication proceeds; the NP **DEDDH exoribonuclease** degrades immunostimulatory RNA, **resulting in** suppression of type I interferon and a **~72-hour delay** in innate antiviral/

inflammatory gene induction (inferred for LUJV from conserved arenaviral biology + macaque transcriptomics) (PMID: 25878103; PMID: 25520505).

5. **BRANCH A (controlled response → survival):** an appropriately timed, measured immune response limits immunopathology → mild/non-lethal disease (macaques; young-female guinea pigs) (PMID: 25520505; PMID: 41118368).
6. **BRANCH B (dysregulated response → death):** delayed-then-explosive **cytokine storm** (IL-1β, IL-2, IL-6, CXCL10) **leads to** systemic immunopathology (PMID: 41118368).
7. Cytokines + **VEGF-mediated endothelial permeability** and endothelial/platelet biochemical dysfunction (loss of prostacyclin, impaired aggregation) **result in** capillary leak, facial/neck edema, and effusions (PMID: 23121052; PMID: 3543155).
8. Hepatocellular injury/necrosis **leads to** deficient synthesis of coagulation factors; combined with consumptive coagulopathy, this **results in DIC** (PMID: 23121052).
9. Capillary leak + DIC + vasoplegia **lead to** distributive/hypovolemic **shock** and **multi-organ failure**, the terminal events in week 2 (PMID: 25393244).

Upstream vs downstream: viral entry/replication and IFN suppression are upstream; cytokine storm, endothelial/platelet dysfunction, and DIC are downstream effectors of tissue injury. **Key GO terms:** viral entry into host cell (GO:0046718), receptor-mediated endocytosis (GO:0006898), fusion of virus membrane with host endosome membrane (GO:0039654), negative regulation of type I interferon production (GO:0032480), inflammatory response (GO:0006954), blood coagulation (GO:0007596), cytokine production (GO:0001816). **Key CL terms:** endothelial cell (CL:0000115), macrophage (CL:0000235), dendritic cell (CL:0000451), hepatocyte (CL:0000182), platelet (CL:0000233), NK cell (CL:0000623). **CHEBI:** ribavirin (CHEBI:63580).

7. Anatomical Structures Affected

- **Primary organs/systems:** vascular endothelium / cardiovascular system (UBERON:0001981 blood vessel endothelium; UBERON:0004535 cardiovascular system); liver (UBERON:0002107); hematologic/coagulation system.
- **Secondary involvement:** kidneys (UBERON:0002113), lungs (UBERON:0002048), spleen (UBERON:0002106) — all show high viral titers in guinea pigs (PMID: 41118368); CNS (late neurological signs); soft tissue of face/neck (edema).
- **Cell level:** endothelial cells, hepatocytes, platelets, mononuclear phagocytes.
- **Subcellular compartments involved in the viral life cycle (GO cellular component):** late endosome (GO:0005770), endosome membrane (GO:0010008), plasma membrane

(GO:0005886), host cell surface. **Localization:** systemic/bilateral; facial and cervical edema is a distinctive regional sign.

8. Temporal Development

Onset: acute, adult-onset, with abrupt fever after an incubation of roughly 1–2 weeks (arenaviral range). **Progression:** week 1 = febrile prodrome and multisystem symptoms; week 2 = shock, DIC, multi-organ failure, death. **Rate:** rapid. **Course:** monophasic, self-limited in survivors (recovery) or fatal. In the one survivor, early ribavirin and intensive support were associated with recovery (PMID: 25393244). The **critical window for intervention** appears to be early, before the week-2 cytokine-storm/DIC cascade.

9. Inheritance and Population

Epidemiology: a single recognized outbreak (2008), 5 cases, **80% CFR (4/5)** (PMID: 27593704). Prevalence/incidence are effectively zero outside this outbreak; true endemic burden is unknown due to under-surveillance in the region (PMID: 25416679). **Inheritance:** not applicable (infectious). **Geographic distribution:** Southern Africa — epidemiologically linked to Zambia (Lusaka) and South Africa (Johannesburg). **Sex/age:** too few human cases for a ratio; experimental models show male sex and adult/older age associated with worse outcomes (PMID: 41118368).

10. Diagnostics

Laboratory: thrombocytopenia, transaminitis, coagulopathy (Lassa-like) (PMID: 29733163). **Molecular diagnosis of choice:** real-time **RT-PCR** assays specific for Lujo viral RNA, developed and validated against the sole isolate (PMID: 24096270). Virus isolation and serology require BSL-4. **BSL-2 surrogate systems** (VSV-LUJV/GP pseudotypes) support research/entry assays (PMID: 24741091; PMID: 22396728). **Differential diagnosis:** Lassa fever (most important — clinically near-identical), Ebola/Marburg, Crimean-Congo hemorrhagic fever, Rift Valley fever, severe malaria, leptospirosis, typhoid, and other septic/DIC syndromes. **Genetic/omics diagnostics:** not applicable for host diagnosis. No newborn/carrier screening applies.

11. Outcome/Prognosis

Mortality: 80% CFR in the only outbreak. **Prognostic factors:** early antiviral therapy and intensive supportive care (favorable — one survivor); magnitude/timing of the inflammatory cytokine response is the principal determinant of outcome, with cytokine storm predicting

death (PMID: 41118368; PMID: 25520505). **Complications:** DIC, hemorrhage, shock, hepatic and multi-organ failure. **Recovery:** survivors can recover; long-term sequelae are undocumented given the tiny cohort. Candidate **prognostic biomarkers:** IL-1β, IL-6, CXCL10 (from models).

12. Treatment

No approved antiviral or vaccine exists (PMID: 41118368). Management is **early ribavirin plus aggressive supportive care** (fluid resuscitation, correction of coagulopathy, organ support), as used in the single survivor (PMID: 25393244) — **NCIT: Ribavirin (C790); Supportive Care (C15277)**. **Experimental/preclinical candidates:**

Strategy	Mechanism / target	Evidence	Status
Ribavirin	Viral RNA synthesis inhibitor	PMID: 25393244	Used clinically (survivor)
CB1 antagonists (rimonabant, AM251, AM281)	Block LUJV entry	PMID: 37633192	Preclinical
FDA-approved library hits	Entry inhibition	PMID: 34925303	Preclinical
Reporter-virus HTS hits (6 compounds)	Various	PMID: 34203149	Preclinical
Griffithsin (lectin)	Binds GP glycans, blocks entry	PMID: 42037406	Preclinical
PF-429242	S1P/SKI-1 inhibitor; blocks GPC maturation	PMID: 26719243	Preclinical

No targeted pharmacogenomics, gene therapy, or cell therapy applies.

13. Prevention

Primary prevention rests on **strict infection control** (barrier nursing, isolation, PPE, safe handling of body fluids), given nosocomial spread (PMID: 27593704; PMID: 29733163). Timely molecular diagnosis enables patient isolation to prevent onward transmission (PMID: 24096270). **Immunization:** no licensed vaccine; the **MOPEVAC** hyperattenuated-Mopeia platform expressing heterologous glycoproteins is a candidate pan-arenavirus vaccine (PMID: 29593043). Reservoir control cannot yet be targeted because the reservoir is unidentified.

14. Other Species / Natural Disease

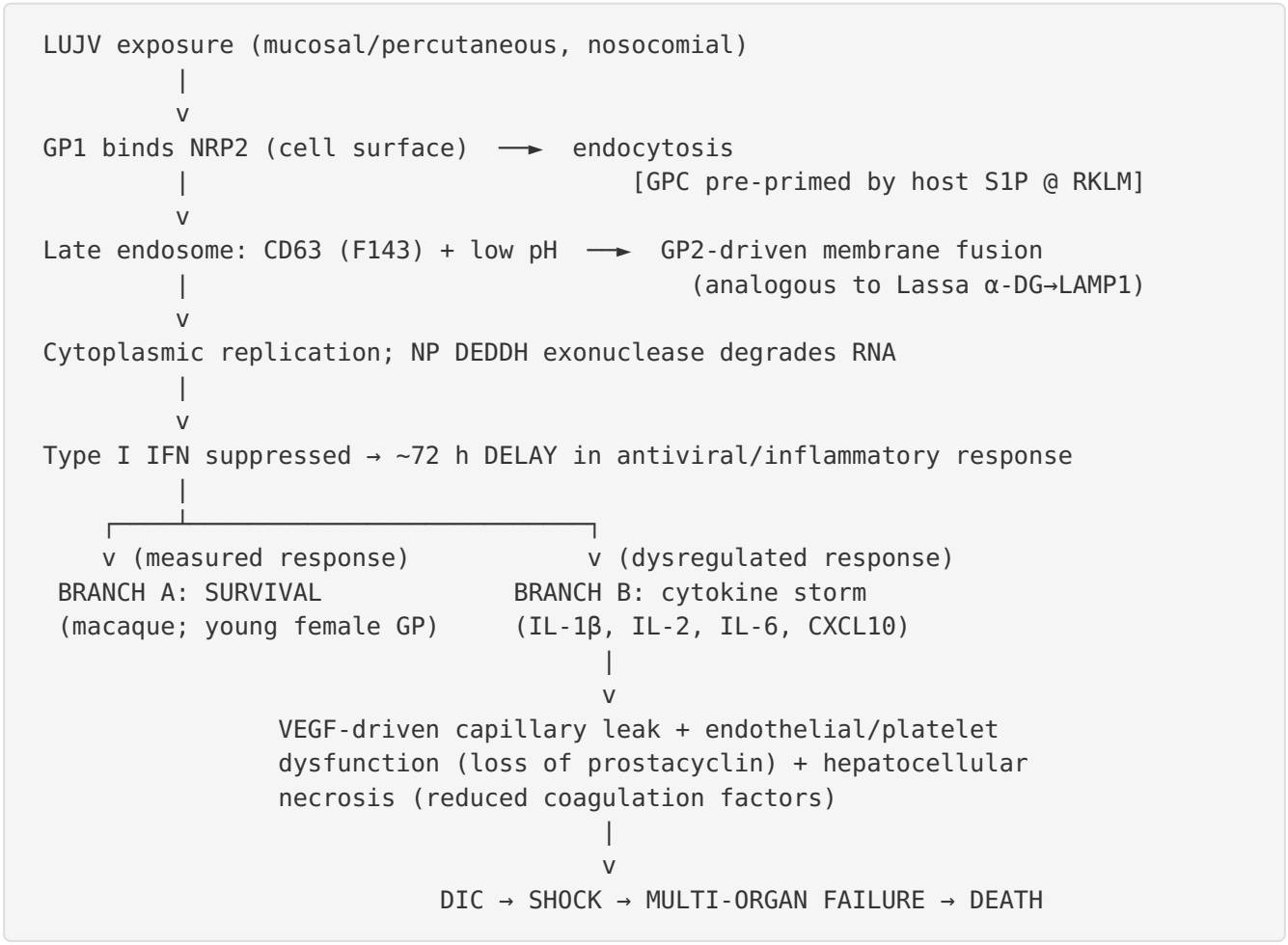
Pathogen: Lujo mammarenavirus (NCBI taxid 649188). **Presumed reservoir:** an unidentified rodent (zoonotic, by analogy to other arenaviruses) — **zoonotic potential is high but the specific host and spillover route are unknown** (PMID: 41118368; PMID: 25416679). No naturally occurring companion-animal or livestock disease has been reported. Experimental cross-species susceptibility is governed by the host CD63 F143 residue (PMID: 35164564).

15. Model Organisms

Model	Type	Recapitulation	Reference
Cynomolgus macaque	NHP	Mild, non-lethal; useful for immune kinetics	PMID: 25520505
Strain 13/N guinea pig	Rodent	Hematologic/biochemical abnormalities, age/sex-dependent lethality	PMID: 41118368
STAT1-knockout mouse	Genetic KO rodent	Partial lethal disease; key hematologic/ clinical-chemistry/histologic features	PMID: 42075721
VSV-LUJV/GP pseudotype	In vitro (BSL-2)	Entry only	PMID: 24741091; PMID: 22396728
Recombinant reporter LUJV	In vitro HTS	Full replication for drug screening	PMID: 34203149

Limitation: wild-type inbred mice/hamsters resist infection (non-human CD63), so humanized-CD63 or IFN-deficient backgrounds are needed; no model perfectly reproduces fatal human LHF.

Mechanistic Model / Interpretation



The central interpretive insight is that **LHF outcome is an immunological phenotype, not a virological one**: viral titers are comparable in survivors and fatalities, but the timing and intensity of the host inflammatory response diverge (PMID: 41118368; PMID: 25520505). This frames two complementary therapeutic avenues: **block entry/replication early** (NRP2/CD63 entry inhibitors, S1P inhibitors, griffithsin, ribavirin) and **modulate the downstream immunopathology** (anti-cytokine/supportive strategies). LUJV's uniqueness among arenaviruses — the NRP2/CD63 receptor pair — makes it both a mechanistic outlier and a sharply defined drug target.

Evidence Base

PMID	Role in this report
19478873	Discovery & phylogeny: LUJV is a distinct Old World arenavirus (Finding 1)
29120745	NRP2 + CD63 host factors; entry mechanism (Finding 2)
35164564	CD63 F143 host-range determinant (Findings 2, 5)
30150732	Crystal structure of GP1–NRP2 (Finding 2)
39169025	Cryo-EM of trimeric spike; single NRP2 at apex (Finding 2)
24741091	Pseudotype entry independent of α-DG/TfR1; NPC1/cholesterol dependence (Findings 2, 5)
25393244	Clinical course & ribavirin in survivor (Finding 3)
29733163	Lassa-like clinical/lab features; outbreak overview (Finding 3)
41118368	Guinea pig model; immune (not viral) burden drives outcome (Findings 4, 5, 6)
25520505	Macaque model; 72-h delayed inflammatory response (Findings 4, 5, 10)
42075721	STAT-1 KO mouse model (Findings 5, 10)
27593704	Epidemiology: 80% CFR, nosocomial (Finding 6)
24096270	RT-PCR diagnostics; single strain (Finding 6)
26719243	S1P maturation at RKLM; ESCRT-independent budding (Finding 7)
37633192	CB1 antagonist entry inhibitors (Finding 8)
34925303	FDA-library entry inhibitors (Finding 8)
34203149	Reporter-virus HTS; 6 anti-LUJV compounds (Finding 8)
42037406	Griffithsin blocks entry (Finding 8)
22396728	BSL-2 pseudotype screening platform (Finding 5)
23121052	Shared VHF tissue-injury mechanisms (Finding 9)
3543155	Platelet/endothelial dysfunction in primate Lassa shock (Finding 9)
29295909	Lassa α-DG→LAMP1 switch paradigm (Finding 9)
25878103	NP DEDDH exonuclease IFN suppression (Finding 10)

PMID	Role in this report
29593043	MOPEVAC vaccine platform (Finding 10)
22607481	Arenavirus drug-discovery context (background)
25416679	Under-surveillance of emerging zoonoses (epidemiology context)

Limitations and Knowledge Gaps

1. **Single outbreak, n=5.** All human clinical knowledge rests on five cases; incidence, prevalence, full phenotype spectrum, sex ratio, incubation period, and long-term sequelae are essentially unknown.
2. **Reservoir unidentified.** No animal host or spillover route is known, precluding reservoir-targeted prevention and ecological risk assessment.
3. **No fully lethal animal model.** Macaques are mild; guinea pig and STAT-1 KO models are only partially lethal; wild-type rodents resist infection due to non-human CD63.
4. **Mechanistic inference by homology.** The NP-exonuclease IFN-suppression step and some VHF tissue-injury mechanisms (VEGF permeability, DIC) are inferred from related arenaviruses rather than demonstrated directly for LUJV.
5. **No approved countermeasures.** All targeted therapeutics and the MOPEVAC vaccine are preclinical; ribavirin efficacy is anecdotal (one survivor who also received intensive supportive care).
6. **Genomic diversity unknown.** Diagnostics are validated against a single isolate; antigenic/genetic variation across any future strains is uncharacterized.

Proposed Follow-up Experiments / Actions

1. **Reservoir surveillance** in Zambia/Southern Africa — trap and screen rodents by RT-PCR/serology to identify the natural host and define spillover ecology.
2. **Develop a robustly lethal small-animal model** — e.g., humanized-CD63 (F143 knock-in) mice or further-characterized guinea pig strains — to enable rigorous efficacy testing.

3. **Test leading entry inhibitors in vivo** — prioritize CB1 antagonists, griffithsin, and the S1P inhibitor PF-429242 in the guinea pig/STAT-1 KO models, alone and with ribavirin.
4. **Directly validate the NP-exonuclease IFN-suppression mechanism in LUJV** by catalytic-residue mutagenesis, testing attenuation and immunogenicity (toward a rationally attenuated vaccine).
5. **Advance MOPEVAC-LUJV** (Mopeia platform expressing LUJV GPC) through immunogenicity and challenge studies.
6. **Define immunopathology biomarkers** (IL-1β, IL-6, CXCL10) as prognostic/therapeutic-response markers and evaluate host-directed immunomodulation to interrupt the cytokine-storm→DIC→shock cascade.
7. **Broaden and field-deploy diagnostics** — expand RT-PCR design to anticipate genetic variants and distinguish LUJV from Lassa in endemic settings.

Report compiled from a 5-iteration autonomous investigation: 10 confirmed findings, 26 papers reviewed. Lujo hemorrhagic fever is an infectious viral hemorrhagic fever with no human causal gene; the causal agent is Lujo virus (LUJV).

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery